

Kirlos Yousef

Senior Software Engineer | AI Application Engineering

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 Egypt | Open to remote work and relocation

Senior software engineer with 5+ years shipping production software and leading engineering teams, specializing in AI application engineering. Built a Python/FastAPI service for structured LLM extraction, hybrid retrieval, and retrieval-backed answers, with reproducible evaluation and CI quality gates. Experienced in product delivery, reliability, and end-to-end ownership across mobile, web, and backend systems

TECHNICAL SKILLS

Backend & Data: Python, TypeScript, FastAPI, Pydantic, Hono, PostgreSQL, pgvector, DynamoDB, OpenSearch

LLM & RAG: OpenAI SDK, OpenRouter, Gemini APIs, structured outputs, tool calling, embeddings, reranking

Cloud & Delivery: AWS Lambda, AppSync, CDK, Cloudflare Workers, Docker / Compose, GitHub Actions

Quality & Evals: pytest, Vitest, Ruff, Pyright, prompt A/B tests, recall@k, MRR, latency / cost tracking

Product & APIs: React, Next.js, Swift, SwiftUI, REST, GraphQL, SSE, OAuth

SELECTED AI PROJECTS

Receipty | LLM Extraction & RAG

- Built a FastAPI vision pipeline with strict JSON schemas, Pydantic validation and explicit review/failure states
- Passed receipt/total/date checks on 60/60 curated images through the shared extraction service
- Built Postgres/pgvector hybrid retrieval, lexical reranking and citation-returning Q&A with not-found handling
- Evaluated four search strategies on 69 labelled queries using recall@k and MRR on queries with gold sources
- Measured 0/60 prediction changes across two v1 repeats; retained v1 after v2 regressed on merchant/currency
- Implemented bounded retries, jitter and deadlines; tracked model settings, prompt/commit hashes, tokens and cost
- Added CI lint/types, an 80% branch coverage gate, dependency audits and unreadable-total regression tests

Revera AI | Voice Transcription , Intelligent Voice Recorder & AI Transcriber

- Built a Whisper/SFSpeech transcription pipeline with offline queuing and recovery from interruptions
- Reduced raw audio upload data by ~82% through 44.1 kHz stereo to 16 kHz mono conversion

Plot My Calories | Vision-based Nutrition App, Sellou project

- Built Plot My Calories end to end, integrating OpenAI Vision for image-based input with asynchronous processing

PROFESSIONAL EXPERIENCE

Product Engineer - AI Applications, Sellou

08/2025 – Present | US (Remote)

Spookling | iOS, web and backend

- Built Spookling's iOS and web apps for AI chat, calendar workflows and browser-based coding
- Integrated SSE streaming and connected LLM tool calls to EventKit calendar actions
- Built workspace/DM chat APIs on AppSync, Lambda and DynamoDB with threads, mentions and file indexes
- Shipped OpenSearch Serverless chat search with indexing, highlights and conversation-level access control
- Implemented Buffer OAuth and Google connection flows; integrated Composio SDK for third-party tools
- Added paginated assistant history and booking APIs; fixed Stripe seats, portal access and credit top-ups

Senior iOS Engineer / iOS Tech Lead, Sellou

02/2025 – 08/2025 | US (Remote)

- Progressed to leading the iOS team; introduced CI/CD workflows that cut release cycle time by 50%
- Cut playback startup 40% with AVKit/HLS and initial download size 20% with On-Demand Resources

iOS Team Lead, Iomob - B2B Mobility-as-a-Service Platform

02/2024 – 02/2025 | Spain (Remote)

- Led 5 iOS engineers delivering LNER and BrightBike while sustaining a crash-free rate above 99.9%
- Added Stripe tokenization to LNER, cutting checkout below 8 seconds and trip-planning latency by 35%

iOS Engineer, Iomob - B2B Mobility-as-a-Service Platform

02/2021 – 02/2024 | Spain (Remote)

- Built mobility SDK used in Ford/Renfe integrations; cut integration time from weeks to under 2 days
- Normalized data from over 10 providers and raised core SDK test coverage above 90% with unit and UI tests

EDUCATION

B.S. Computer Science, University of Debrecen

2017 – 2020 | Debrecen, Hungary

- Graduated first in cohort | Fully funded scholarship